INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY
HYDERABAD

Research Teaser for IoT'25: Software Task 1

Submission Guidelines:

1. Submit your code along with a report detailing your implementation, analysis, and findings.
2. Include all plots and their interpretations in the report.
3. Clearly explain the optimization technique you chose and provide benchmarks comparing the optimized version with the original implementation.
4. Ensure your code is well-commented and follows good programming practices.
5. Please submit a single zip folder named StudentName_College_SWTask1.zip containing StudentName_College_report.pdf and StudentName_College_recommender.(ipynb or py) and a README.md file

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1 Import Dataset into PostgreSQL & MongoDB with Vertical-Based Tables/Collections

1.1 Objective

The goal of this task is to practice reading datasets, creating database tables dynamically, and inserting data into PostgreSQL & MongoDB based on different categories (verticals) in the dataset.

1.2 Instructions

1.2.1 Dataset Overview

You are provided with a dataset file. The dataset contains multiple types of data, each categorized under a vertical. Each vertical may have different columns specific to its type.

1.2.2 Read the Dataset

- Use Python to read the dataset file.
- Explore the dataset to understand its structure, columns, and verticals.

1.2.3 Step 2: Create PostgreSQL Tables & MongoDB collections

- For each unique vertical in the dataset, create a separate table & collection in PostgreSQL & MongoDB
- Ensure that the column names and data types in each table match the dataset.
- Define appropriate primary keys for each table.

1.2.4 Insert Dataset into Tables

- Handle missing values and ensure data integrity.
- Insert the rows corresponding to each vertical into the respective table.

1.3 Deliverables

- Python script to read the dataset file and insert data into PostgreSQL & MongoDB.
- output showing that the data is successfully inserted.

2 Build Machine Learning Models for Each Vertical

2.1 Objective

The goal of this task is to use the dataset from Task 1, split it by verticals, and create a separate machine learning model for each vertical. This will help you practice data preprocessing, model training, and evaluation for different categories within the same dataset.

2.2 Instructions

2.2.1 Extract Data by Vertical

- You can either utilize the dataset imported into PostgreSQL in **Task 1** or simply use the dataset file directly.
- Understand the features and target variable for each vertical.

2.2.2 Preprocess Data

- Clean the data for each vertical (handle missing values, outliers, redundant columns).
- Encode categorical variables and scale/normalize numerical features as needed.
- Split each vertical's data into training and testing sets.

2.2.3 Build Machine Learning Model per Vertical

- Select a suitable model for each vertical(all three models should be used).
- Train the model on the training set and evaluate its performance on the testing set.
- Hypertune the model as necessary.

2.2.4 Evaluate and Document Results

- Use appropriate metrics for each model.
- Summarize results, observations, and insights for each vertical.

2.3 Deliverables

A Python script or notebook for each vertical including:

- Data preprocessing steps.
- Model training and evaluation.
- Performance metrics and observations.

3 Improve Model Accuracy with Additional Data (Bonus Task)

3.1 Objective

The goal of this task is to enhance the machine learning models created in Task 2 by collecting additional data and improving model accuracy. This will help you practice data augmentation, feature engineering, and model optimization.

3.2 Instructions

3.2.1 Collect Additional Data

- Each student should generate or collect additional data points relevant to the dataset from Task 1.
- The new data should be realistic and aligned with the features in the existing dataset.

3.2.2 Integrate New Data

- Combine the new data with the original vertical datasets from Task 2.
- Ensure data consistency and handle any missing or inconsistent values.

3.2.3 Retrain Models

- Retrain each vertical's model using the expanded dataset.
- Explore feature engineering, new variables, or data transformations to improve performance.

3.2.4 Optimize Models

- Tune hyperparameters to improve accuracy or other relevant metrics.
- Consider ensemble methods or advanced models if applicable.

3.2.5 Evaluate and Compare

- Evaluate the retrained models using the same metrics as Task 2.
- Compare the results with the previous models and document improvements.

3.3 Deliverables

Updated Python script or notebook for each vertical:

- Data integration and preprocessing.
- Retrained and optimized model.
- Performance evaluation and comparison with previous results.